

Jan Sobotka

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EDUCATION

- Swiss Federal Institute of Technology in Lausanne (EPFL)** Sep. 2024 – ongoing
M.S. in Computer Science Lausanne, CH
- Weighted grade average: **5.89** (scale of 1.0 to 6.0, 6.0 is the highest)
 - Anticipated graduation date: Aug. 2026
- Czech Technical University, Faculty of Information Technology** July 2021 – June 2024
B.S. in Informatics, Specialization in Artificial Intelligence Prague, CZ
- Weighted grade average: **1.05** (scale of 1.0 to 4.0, 1.0 is the highest)

RESEARCH EXPERIENCE

- Research Assistant at the Autonomous Systems Group** July 2025 – ongoing
University of Texas at Austin, Oden Institute for Computational Engineering and Sciences Austin, US
- Analyzing large language models in strategy games using mechanistic interpretability and game theory.
 - Supervised by Prof. Ufuk Topcu.
- Research Assistant at the MLBio Lab** Aug. 2024 – July 2025
Swiss Federal Institute of Technology in Lausanne (EPFL) Lausanne, CH
- Enhanced *Weak-to-Strong generalization* (learning from weaker teachers) by dynamically reweighting teaching signals, improving robustness under distribution shifts by over 30% [2].
 - Evaluated in-context learning capabilities of multimodal foundation models on computer vision tasks.
 - Supervised by Prof. Maria Brbić.
- Research Intern at the Computational Systems Neuroscience Group** Sep. 2023 – Aug. 2024
Faculty of Mathematics and Physics, Charles University Prague, CZ
- Developed a state-of-the-art method for decoding images from neural population activity [1].
 - Supervised by Mgr. Ján Antolík, Ph.D.
- Computational Neuroscience Research Intern** July 2023 – Sep. 2023
Biozentrum, University of Basel Basel, CH
- Designed computational models of spiking neural networks and applied dynamical systems theory to demonstrate how bistable neuronal dendrites enable a 1,000-fold increase in memory capacity.
- Research Assistant at the Data Science Lab** Apr. 2023 – Feb. 2024
Faculty of Information Technology, Czech Technical University Prague, CZ
- Analyzed training dynamics of *Learning-to-Optimize* (meta-learning) [6,7] and *fractional gradient descent* [5].

PUBLICATIONS

* DENOTES EQUAL CONTRIBUTION

- [1] MEIcoder: Decoding Visual Stimuli from Neural Activity by Leveraging Most Exciting Inputs. [PDF 📄]
Jan Sobotka, Luca Baroni, Ján Antolík.
Neural Information Processing Systems (NeurIPS 2025).
- [2] Weak-to-Strong Generalization under Distribution Shifts. [PDF 📄]
Myeongho Jeon*, **Jan Sobotka***, Suhwan Choi*, Maria Brbić.
Neural Information Processing Systems (NeurIPS 2025).
- [3] Reverse-Engineering Memory in DreamerV3: From Sparse Representations to Functional Circuits. [PDF 📄]
Jan Sobotka, Auke Ijspeert, Guillaume Bellegarda.
Neural Information Processing Systems (NeurIPS 2025, Spotlight at Mech Interp Workshop).

- [4] Do LLMs Strategically Reveal, Conceal, and Infer Information? A Theoretical and Empirical Analysis in The Chameleon Game. [PDF 
Mustafa O. Karabag, **Jan Sobotka**, Ufuk Topcu.
Under review (2025).
- [5] Enhancing Fractional Gradient Descent with Learned Optimizers. [PDF 
Jan Sobotka, Petr Šimánek, Pavel Kordík.
ArXiv preprint (2025).
- [6] Investigation into the Training Dynamics of Learned Optimizers. [PDF 
Jan Sobotka, Petr Šimánek, Daniel Vašata.
International Conference on Agents and Artificial Intelligence (ICAART 2024).
- [7] Investigation into the Training Dynamics of Learned Optimizers (Student Abstract). [PDF 
Jan Sobotka, Petr Šimánek.
AAAI Conference on Artificial Intelligence (AAAI 2024).

INDUSTRY EXPERIENCE

- AI/ML Engineer Junior** June 2021 – June 2023
Generali Česká pojišťovna Prague, CZ
- Prepared a machine learning pipeline that improved the accuracy of product recommendations by 35%.
 - Built computing infrastructure for the company's internal data science community with over 90 members.
 - Applied deep learning to product and text recommendation, client departure prediction, email classification, and unsupervised customer segmentation.
- IT Generalist** Feb. 2020 – June 2021
Startup Disrupt Prague, CZ
- Led the development of a web application for ticket sales.
 - Managed IT setup and support for over 30 offline and online events.

SELECTED PROJECTS

- Deep Reinforcement Learning for Optimal Experimental Design in Biology** Jan. 2023 – June 2023
• Open-science project focused on efficient estimation of biological system parameters [OpenBioML .
- Generative Models of Regulatory DNA Sequences Based on Diffusion Models** July 2022 – Dec. 2022
• Open-science project investigating the application of diffusion models to genomics data [OpenBioML .

EXTRACURRICULAR ACTIVITIES

- Organizer of the Traion Community of Student Entrepreneurs** June 2020 – Feb. 2021
• Organized offline meetings, educational seminars, and workshops for student entrepreneurs.
- Volunteer for an Entrepreneurship Education Program for Students** Dec. 2019 – Aug. 2020
• Organized events and wrote a technology/entrepreneurship blog for the Soutěž and Podnikaj organization.
- Pitcher at the Czech Republic National Baseball Team U-15** Jan. 2017 – July 2017
• Secured third place at the U-15 European Baseball Championship 2017.

HONORS AND AWARDS

- The Bakala Foundation Scholarship:** Awarded to top 7% of applicants (academics & extracurriculars) | 2024
Merit-Based Scholarship, Czech Technical University: Awarded for a high academic average | 2021, 2022, 2023
National Benchmark Exam in Mathematics SCIO : Scored higher than 97% of the 875 test takers | 2021
Algorithms & Programming Competition FIKS: Ranked 4th out of 107 participants | 2020
TOP25 Czech High School Students of the Year 2020: Selected based on extracurricular activities | 2020

SKILLS

- Programming languages:** Python, C, C++, Julia, JavaScript, Go
Other selected technologies: PyTorch, Scikit-learn, NumPy, Pandas, Matplotlib, Azure, AWS, Git, Docker, Bash
Languages: Czech (native), English (C1, TOEFL iBT 105), German (A2)